Alex Key

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Objective:

A curious undergraduate with interests in composite design and manufacturing, structural analysis, and streamlining R&D through modern computer-aided manufacturing technologies. Seeking an internship in launch vehicle structures, propulsion, or testing with opportunities for initiative-driven project work.

Education, Certifications, & Honors:

North Carolina State University, Raleigh, NC *Anticipated May 2027*

Bachelor of Science in Aerospace Engineering | Minor in Computer Programming | GPA: 4.0

Relevant Coursework: Foundations of Graphics, Thermal-Fluid Sciences, Aerodynamics I & II, Fundamentals of Vibrations, Applied Finite Element Analysis, Aerospace Structures I & II, Principles of Automatic Control, C & Software Tools, Software Development Fundamentals, Data Structures

2025 Brooke Owens Fellow

Certifications: Level 2 Certification for the Tripoli Rocket Association, Level 1 FreeFlyer Certification

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Skills

- **CAD:** Solidworks *(Associate Certification)*, Fusion360, Onshape, Meshgrid, and Aspire
- **CAM:** Glowforge/ULS Laser Cutters, Bambu/Flashforge 3D Printers, Shopbot CNC Machines
- **Finite Element Analysis:** ANSYS Workbench, ANSYS APDL, Solidworks Simulation Tool
- **Coding Languages:** Java, MATLAB, Python, C, C++, Assembly, JavaScript, HTML/CSS

Projects

Mach 3 Rocket | *NCSU HPRC Experimental Team* *October 2025 -*

- Designed a vehicle in OpenRocket to reach Mach 3 with strict flight ceiling limitations
- Designed & manufactured a custom length Von Kármán fiberglass nose cone
- Refined a two-part nosecone mold for better composite fiber ratio and more consistent dimensions
- Designed an avionics bay in SolidWorks that minimized weight and fit airframe/space constraints

Experimental Solid-Propellant Rocket Motor *March 2025 -*

- Researched the effects of grain/nozzle geometry and molecular composition on specific impulse and thrust
- Designed a composite motor configuration targeting a specified thrust & burn time using openMotor
- Manufactured a small composite motor with a local expert using a homemade propellant recipe

First-Year Engineering Design Day Arcade Machine | *1st Place* *January 2024 - April 2024*

- Constructed a small arcade machine with a spring-loaded launcher and electronic scoring system
- Designed and toleranced an assembly in SolidWorks of the housing, launcher, and the ball return system
- Created a wooden housing using a Shopbot Desktop Max and powered saws

Sundial Furniture Project | *Bowman Brockman Endowment* *October 2022 - May 2023*

- Designed and created tables and chairs for the sundial in the center of campus in Fusion360 and Onshape
- Cut the furniture out of PVC on a PRS5 Alpha ShopBot CNC machine

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Experience:

NCSU High Powered Rocketry Club | *Member, Vice President (2025 -)* *August 2023 -*

- Designed & optimized vehicle designs in OpenRocket for improved speed & safety margins

- Designed & analyzed subsystems to improve vehicle modularity with SolidWorks & ANSYS Workbench
- Manufactured composite molds and electronics housings with appropriate tolerances using 3D printers
- Manufactured three-dimensional flight hardware using a Shopbot Desktop Max
- Taught the experimental team (~30 people) composite manufacturing techniques & CNC machining
- Designed in-house composites using a self-made MATLAB program to optimize fiber orientation
- Tested in-house manufactured composites for strength and laminate integrity using a Universal Testing Machine
- Designed recovery systems with appropriate shock cord lengths and modified chute release hardware
- Analyzed accelerometer and barometer-based flight data to validate simulation results
- Coordinated with electronic subteams to efficiently integrate payloads and flight control systems
- Organized launch events and drafted checklists for thorough launch day preparations
- Edited formal documentation drafts for the team's NASA Student Launch Competition PDR & CDR

NCSU Additive Manufacturing Lab | ***Gas Atomizer Research Assistant*** *October 2024 -*

- Designed and manufactured safety equipment parts for an Arcast gas atomizer
- Integrated a Pureaire oxygen sensor into an automated data collection system through a Nanodac controller
- Designed and manufactured molds for ceramic nozzles based on technical drawings from a commercial seller
- Analyzed nozzle performance through yield analyses and adjusted geometry for optimized performance
- Performed material analysis on various in-house manufactured powdered metals with ONH & PSD analyzers
- Analyzed defects in L-PBF-manufactured MEA samples to optimize gas atomizer system characteristics

a.i. Solutions | ***2025 Summer Software Engineering Intern*** *May 2025 - August 2025*

- Developed supplemental programs to read CCSDS orbital data messages using Python and FreeFlyer
- Debugged FreeFlyer scripting language and implemented solutions to reduce technical debt using C++
- Managed project tasks and related issues through Jira ticketing system

NCSSM High Powered Rocketry Club | ***Member*** *September 2021 - April 2023*

- Manufactured bulkheads, housings, and other parts using CNC machining, 3D printing, and power tools
- Strengthened vehicle structures with composite layup techniques

NCSSM Fabrication Lab | ***Teaching Assistant*** *August 2022 - May 2023*

- Used CNC machines to manufacture large parts for student/staff projects, such as bike parts and prom statues
- Trained students on CAM equipment such as the laser cutters, 3d printers, and sticker makers
- Designed and printed replacement parts and housing for broken shop & CAM machines
- Guided students in using TinkerCAD & Fusion360 to design 3D prints for personal projects

NCSSM Summer Research & Innovation Program (SRIP) | ***Research in Physics*** *June 2022 to July 2022*

- Designed a simulation in Python, predicting the growth rate of the universe based on the Friedmann equations
- Modified the equations used in the simulation to substitute dark energy with a theoretical negative mass
- Animated a brief video summarizing the findings, communicating in terms understandable to the public